Boundaries and 'Not Sure'

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Boundaries and 'Not Sure' before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Save New Examples for Testing

Without opening the old lesson, explain one key idea from Save New Examples for Testing. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Learning from Examples

Without opening the old lesson, explain one key idea from Learning from Examples. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

If/Then Decisions

Without opening the old lesson, explain one key idea from If/Then Decisions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Boundaries and 'Not Sure' trace and record the first step you can explain without help.

Use W2: Keep a stable **prediction** within its limits for **Uncertainty** and review. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Boundaries and 'Not Sure' explanation against the supplied facts before discussing it.

Use the worked case for **Uncertainty** and review. Proposed explanation: “Two possible values imply equal probabilities”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Transfer the method to distance

A paper model predicts far at a distance of at least 4 units, otherwise near, for whole-number distances 1 through 8. For each complete possibility set, compare **prediction**s: 2 or 3; 3 or 4; 4 or 5.

Identify the set requiring review before a single category can be stated. Design a relevant check and an **unresolved** ending. Limit the claims for the other two sets.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Boundaries and 'Not Sure' target to a trace and a justified conclusion.

A sorter receives a clear purple card, a red card hidden by a sleeve and a card labelled both red and blue by two reviewers. Give a different reason for **uncertainty** in each case and a suitable next action.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Boundaries and 'Not Sure' case and show the deciding evidence.

A sorter handles clearly red and clearly blue cards. Purple and very dark cards arrive.

> How should it handle each uncertain case?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A sorter knows pencils and books. You show a lunch box. Should it force a known label or ask for help? State a useful fallback.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, **boundary case** or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Boundaries and 'Not Sure' adds to Helpful Classroom Sorter and keep the evidence.

Add a not-sure branch, missing-information prompt, and human review destination to the sorter.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A sorter receives a clear purple card, a red card hidden by a sleeve and a card labelled both red and blue by two reviewers. Give a different reason for **uncertainty** in each case and a suitable next action.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

W1 Describe the uncertainty precisely

> Describe the uncertainty precisely. Apply the supplied Boundaries and 'Not Sure' facts and keep your reasoning.

For a smudged height that could be 5 or 6 cm, state what is known, what is unknown and the two **prediction**s under tall at least 6. Choose a check that could distinguish the heights.

Explain why rerunning the threshold comparison cannot do that.

My reasoning and evidence

W2 Keep a stable prediction within its limits

> Keep a stable prediction within its limits. Apply the supplied Boundaries and 'Not Sure' facts and keep your reasoning.

Assume the only possible heights are 8 and 9 cm. Apply tall at least 6 to each. Write a justified result statement and two things it does not establish. Do not invent a confidence percentage.

My reasoning and evidence

W3 Choose useful review evidence

> Choose useful review evidence. Apply the supplied Boundaries and 'Not Sure' facts and keep your reasoning.

A record says 60 but omits the unit. A reviewer suggests reading it as 60 cm because the model needs cm. Explain the problem, name the **evidence** to request and state the outcome if the unit cannot be established.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Transfer the method to distance

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Transfer the method to distance. Apply the supplied Boundaries and 'Not Sure' facts and keep your reasoning.

A paper model predicts far at a distance of at least 4 units, otherwise near, for whole-number distances 1 through 8. For each complete possibility set, compare **prediction**s: 2 or 3; 3 or 4; 4 or 5.

Identify the set requiring review before a single category can be stated. Design a relevant check and an **unresolved** ending. Limit the claims for the other two sets.

> **My reasoning and evidence**

WORKBOOK · 16 OF 21

W5 Review the reviewer’s conclusion

> Review the reviewer’s conclusion. Apply the supplied Boundaries and 'Not Sure' facts and keep your reasoning.

A reviewer cannot read the original tower card but chooses 6 cm because two models both predict tall when given 6. Explain why this fails to resolve the **input**. Write a more defensible review record and a next step.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Boundaries and 'Not Sure'

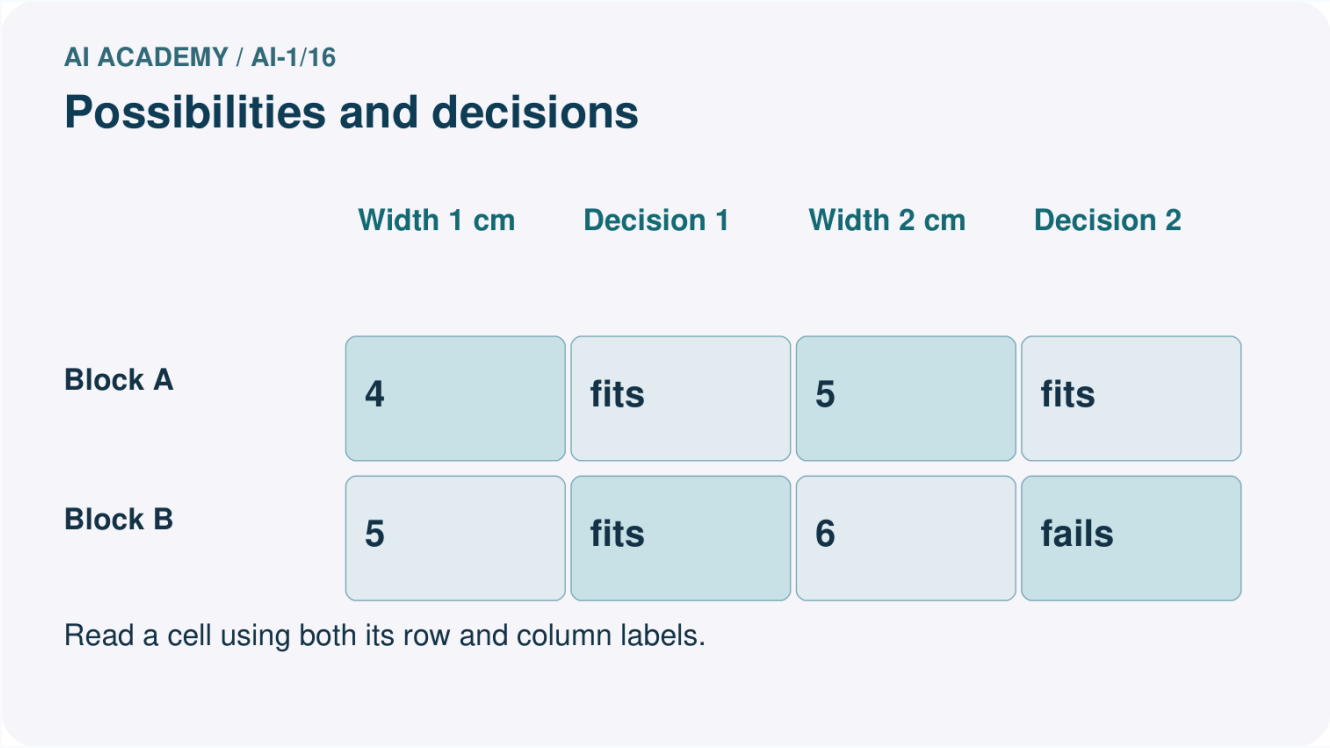
> A fictional cubby selector accepts a block if width is at most 5 cm. Explain the labelled step

> before checking the changed condition.

A fictional cubby selector accepts a block if width is at most 5 cm.

An unclear measurement means Block A is either 4 or 5 cm; Block B is either 5 or 6 cm.

These complete possibility sets contain all widths under consideration. Other dimensions are known to fit.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/16

Boundaries and 'Not Sure'

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Block A possibilities

4 and 5 both satisfy the width limit, which includes equality.



2. Block B possibilities

5 fits; 6 does not.



3. Compare decisions

A has one supported decision; B's possible decisions disagree.



4. Review B

Ask a named reviewer to remeasure B using the specified width unit.

CHANGE ONE THING

Narrow B's possibilities to 6 or 7 cm.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Which original block needs a better width before accepting or rejecting it? What should the reviewer do?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Add explicit review paths to the sorter, including **unresolved input**, responsible person and **evidence** needed.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Helpful Classroom Sorter record.

Project: Helpful Classroom Sorter

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Add explicit review paths to the sorter, including **unresolved input**, responsible person and **evidence** needed.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional cubby selector accepts a block if width is at most 5 cm.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Ask a named reviewer to remeasure B using the specified width unit.

1. Block B possibilities

2. Compare decisions

3. Block A possibilities

4. Review B

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Review B Ask a named reviewer to remeasure B using the specified width unit. **Uncertainty** about a number need not always create **uncertainty** about the decision.

Every allowed width for A fits, so its width **uncertainty** does not affect acceptance. B straddles the boundary.

Record both possibilities and request a better measurement instead of selecting the one that makes a preferred outcome.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Narrow B's possibilities to 6 or 7 cm. Which conclusion is supported by the supplied facts?

1. Both fail the width limit; rejection no longer depends on resolving their difference.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Both fail the width limit; rejection no longer depends on resolving their difference. All remaining possibilities lead to the same decision.

Repair a missing building block

Review Save New Examples for Testing (ai-1/15). Goal: Separate examples used to choose a model from examples reserved for its final evaluation. Short practice: Does 4/4 on the replay replace the original 3/4 result? Explain..

Return to this week and explain what you changed.

Review Learning from Examples (ai-1/13). Goal: Fit a threshold by tracing **prediction**s and counting disagreements for every candidate. Short practice: Which new measured length would reveal disagreement between the candidates?.

Return to this week and explain what you changed.

Review If/Then Decisions (ai-1/10). Goal: Evaluate a condition using the current state and trace only the selected branch. Short practice: Which **input** detects the difference between the two written conditions?

Explain its entire run.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

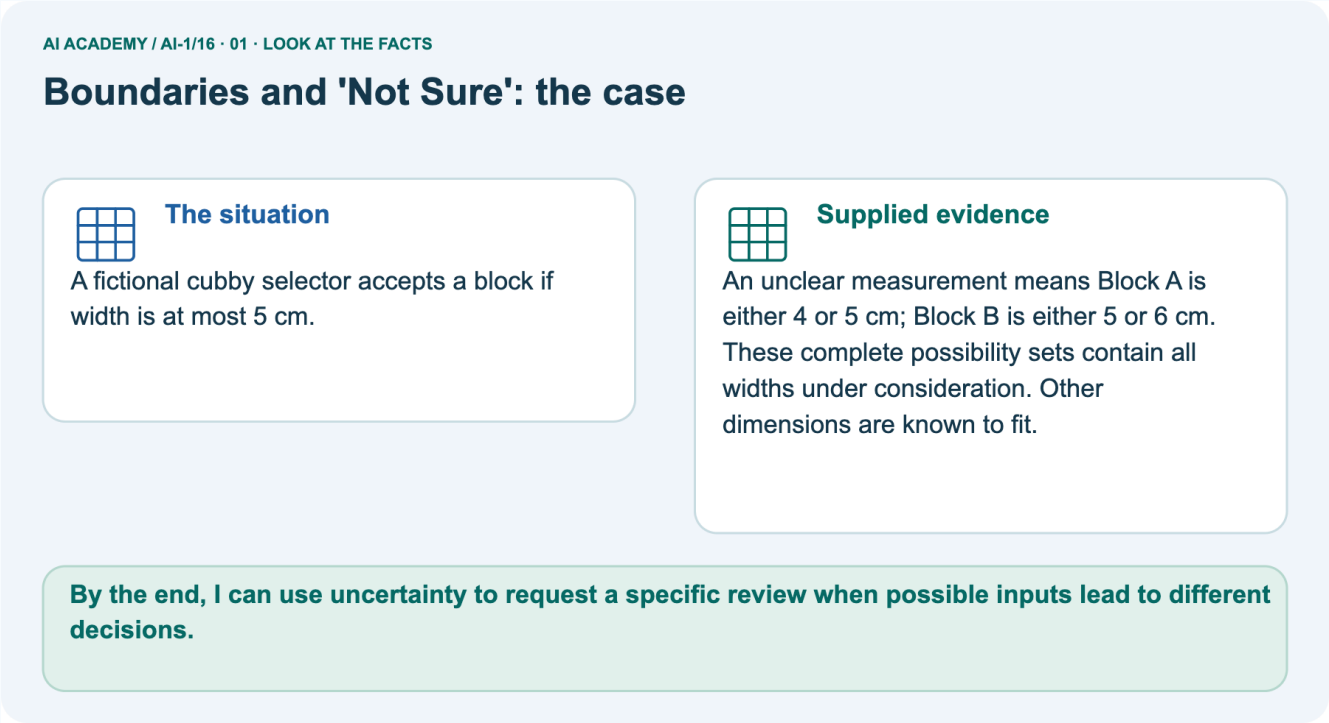
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Boundaries and 'Not Sure' picture from its labels, then record a relevant limit.

01. Boundaries and 'Not Sure': the case



Follow the labelled picture with the matching explanation. Open the vector diagram

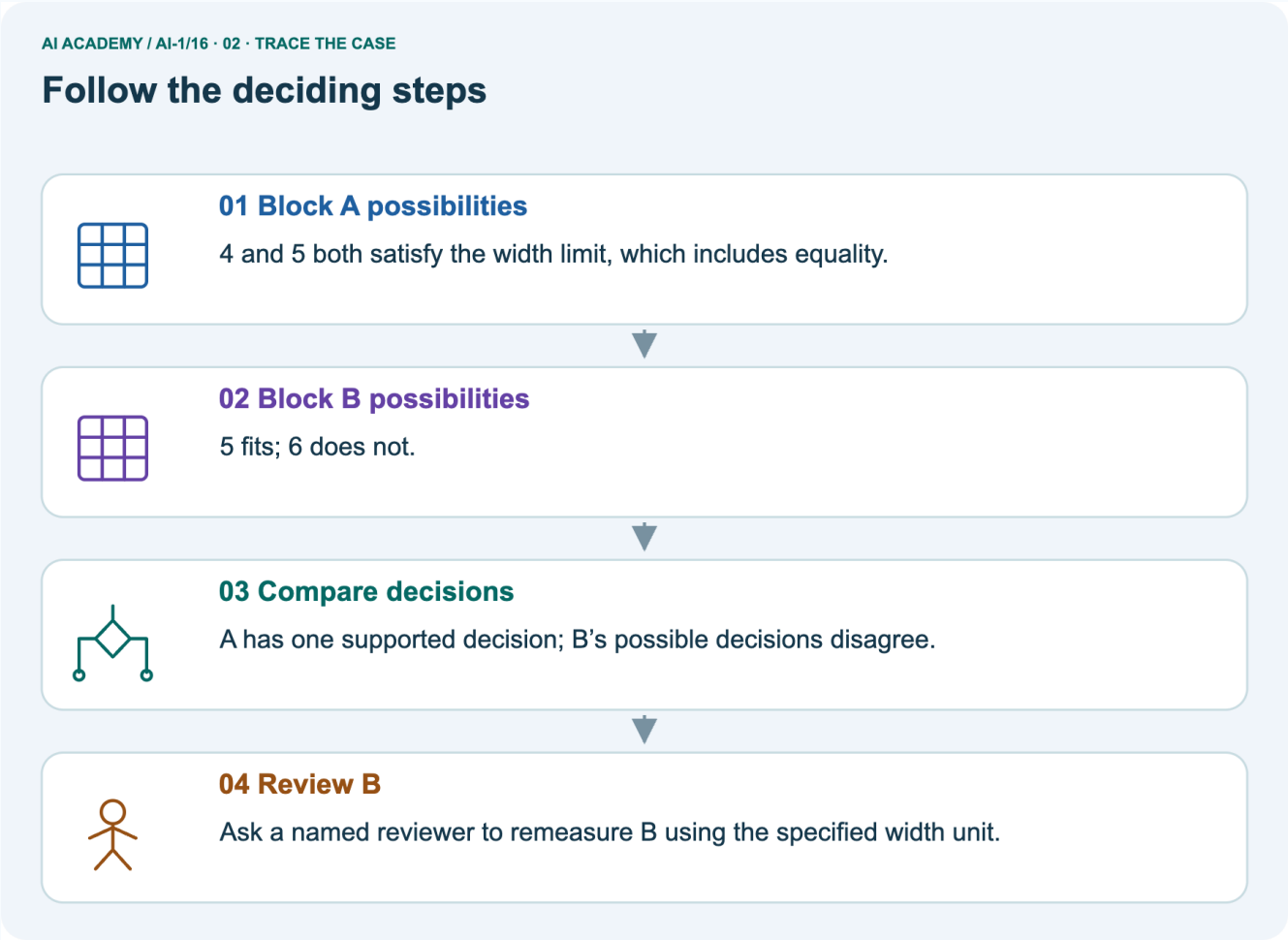
A fictional cubby selector accepts a block if width is at most 5 cm. Supplied facts: An unclear measurement means Block A is either 4 or 5 cm; Block B is either 5 or 6 cm.

These complete possibility sets contain all widths under consideration. Other dimensions are known to fit.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



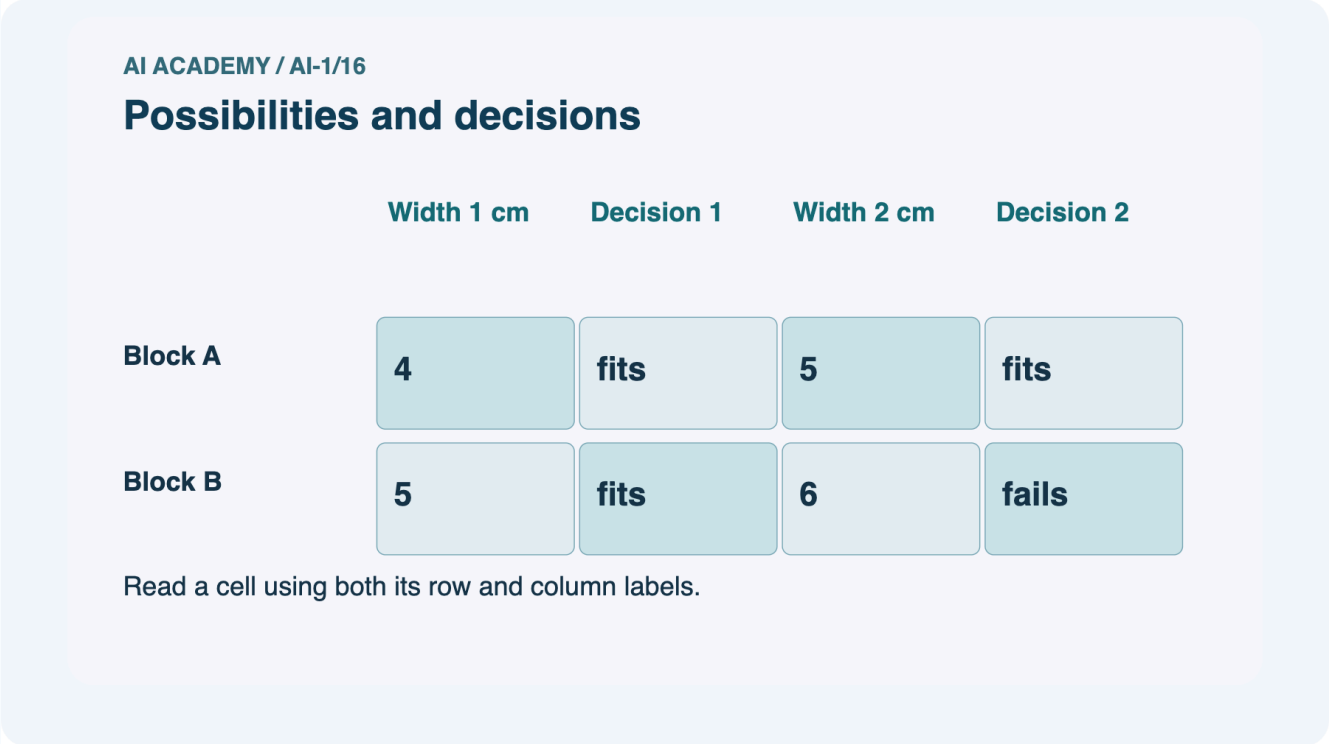
Follow the labelled picture with the matching explanation. Open the vector diagram

Block A possibilities: 4 and 5 both satisfy the width limit, which includes equality. → Block B possibilities: 5 fits; 6 does not. → Compare decisions: A has one supported decision; B's possible decisions disagree. → Review B: Ask a named reviewer to remeasure B using the specified width unit.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Possibilities and decisions



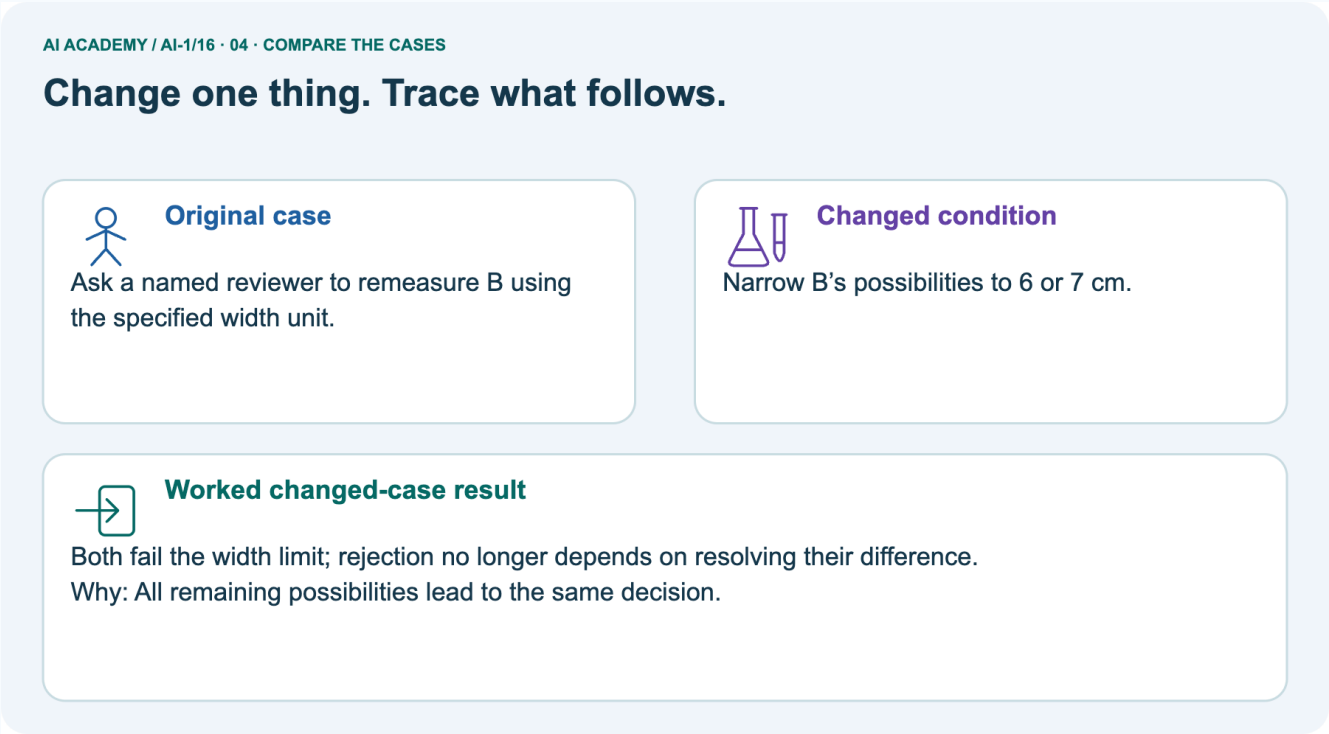
Follow the labelled picture with the matching explanation. Open the vector diagram

Possibilities and decisions. Block A: Width 1 cm = 4, Decision 1 = fits, Width 2 cm = 5, Decision 2 = fits. Block B: Width 1 cm = 5, Decision 1 = fits, Width 2 cm = 6, Decision 2 = fails.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

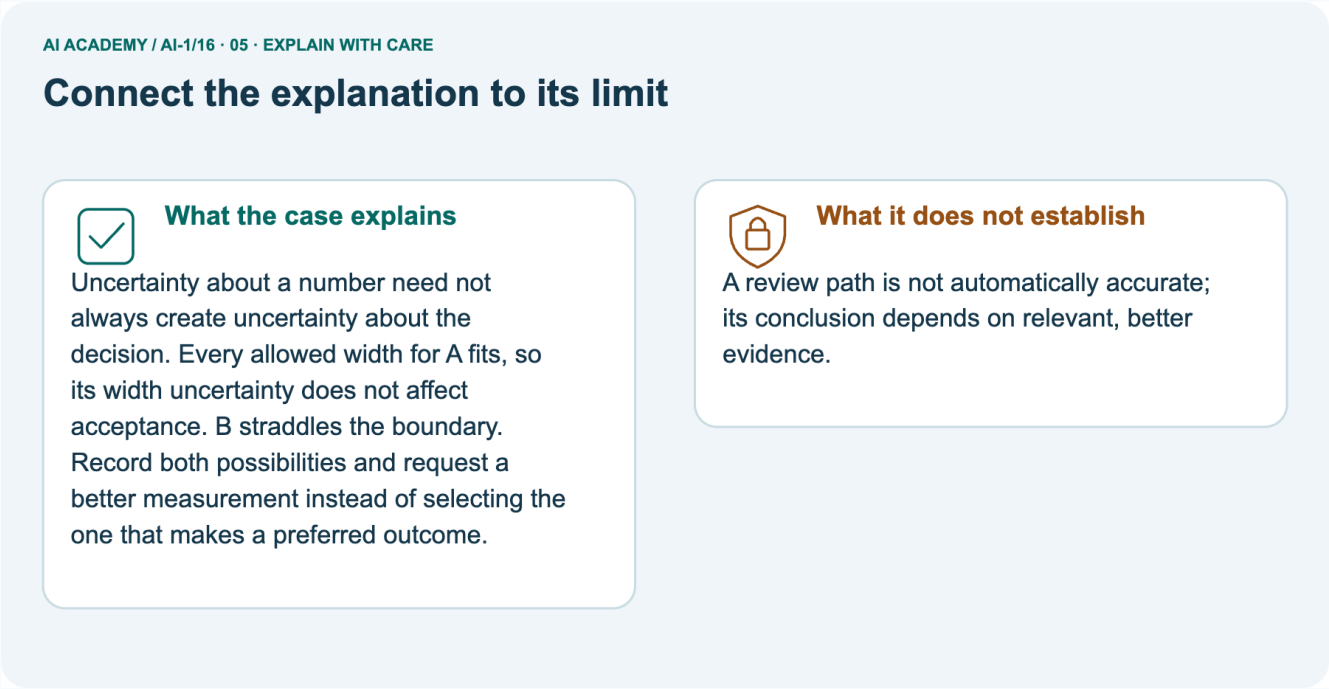
Original case: Ask a named reviewer to remeasure B using the specified width unit. Changed condition: Narrow B's possibilities to 6 or 7 cm.

Expected result: Both fail the width limit; rejection no longer depends on resolving their difference. Reason: All remaining possibilities lead to the same decision.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

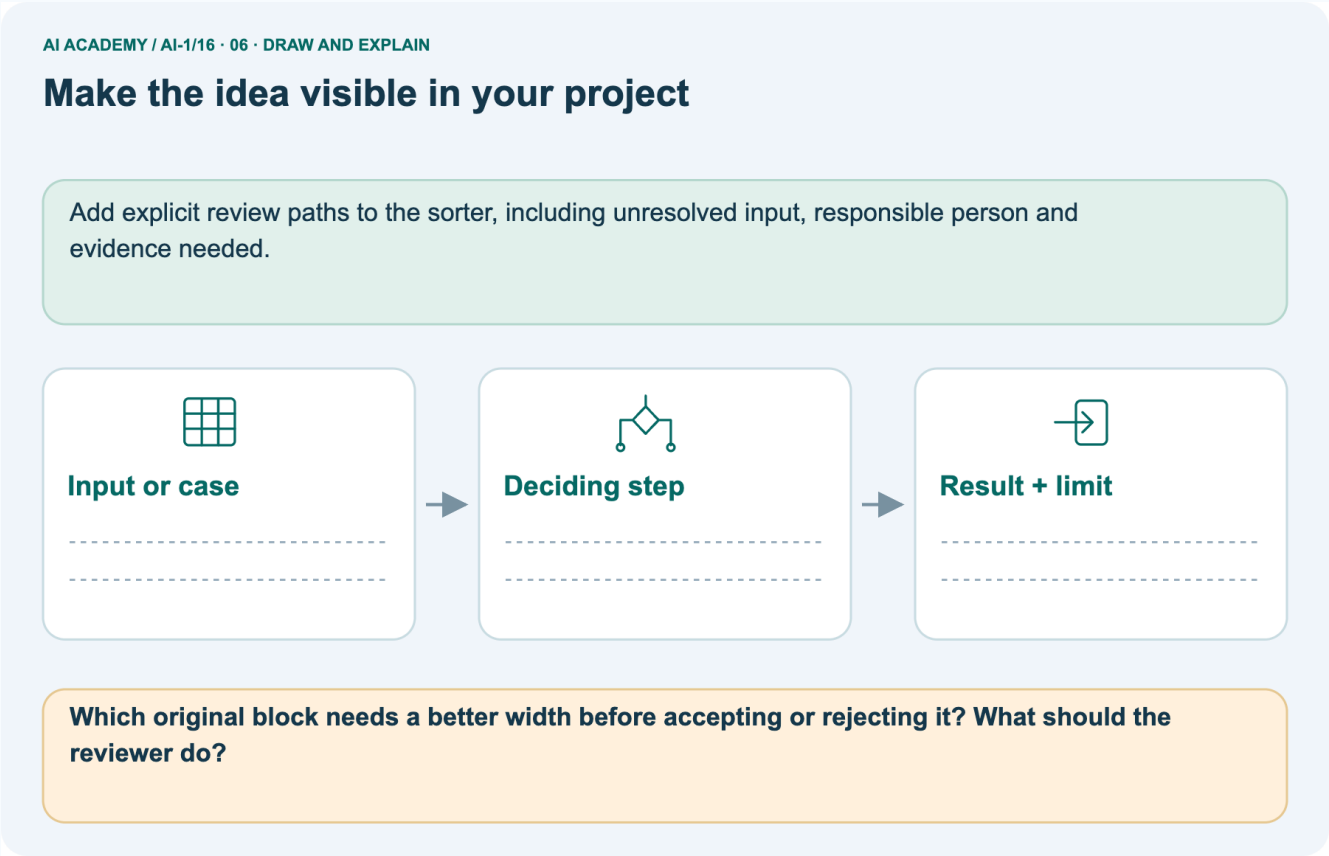
Uncertainty about a number need not always create **uncertainty** about the decision. Every allowed width for A fits, so its width **uncertainty** does not affect acceptance. B straddles the boundary.

Record both possibilities and request a better measurement instead of selecting the one that makes a preferred outcome. Limit: A review path is not automatically accurate; its conclusion depends on relevant, better **evidence**.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Add explicit review paths to the sorter, including **unresolved input**, responsible person and **evidence** needed. Draw your **input**, deciding step, result and limit.

Supported question: Which original block needs a better width before accepting or rejecting it? What should the reviewer do?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.